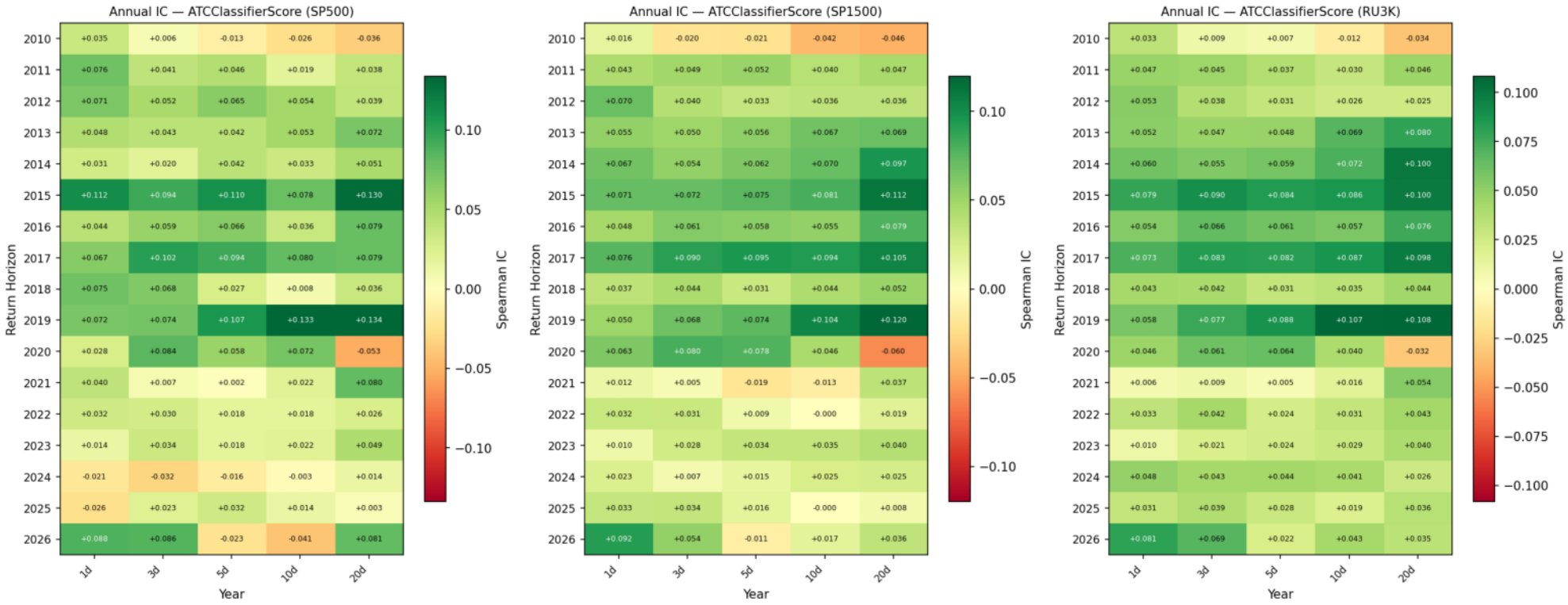
Backtesting the ProntoNLP Earnings-Call ATC Signal

Backtest Charts Bundle

Yueqi Lin · May 2026

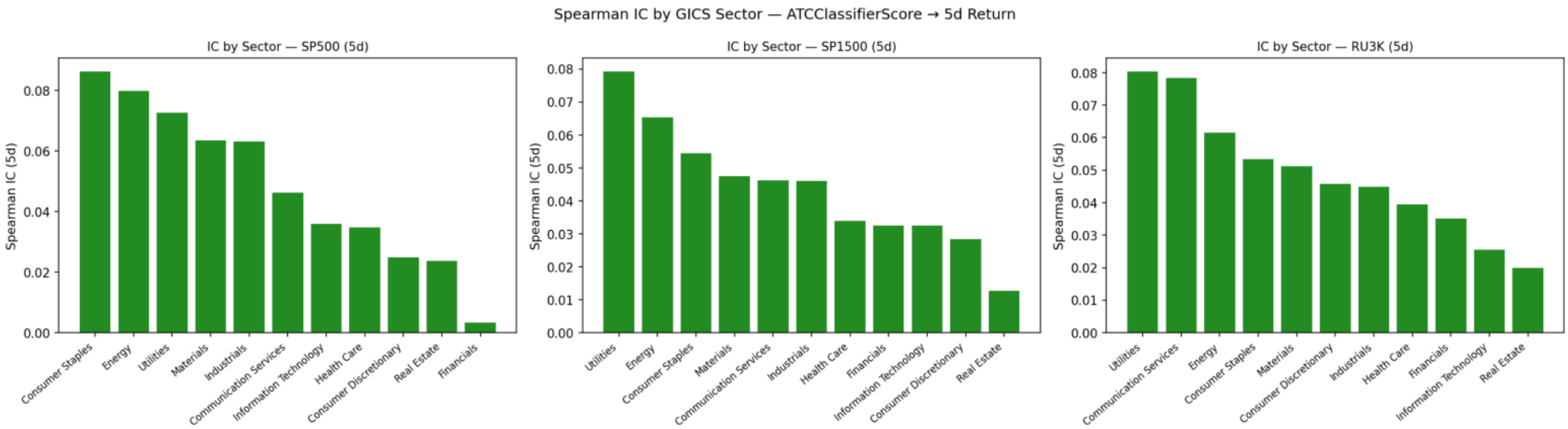
21 figures · S&P 500 / S&P 1500 / Russell 3000
5 bps one-way TC · Expanding walk-forward 2018Q1–2026Q2

§1a Annual IC Heatmap



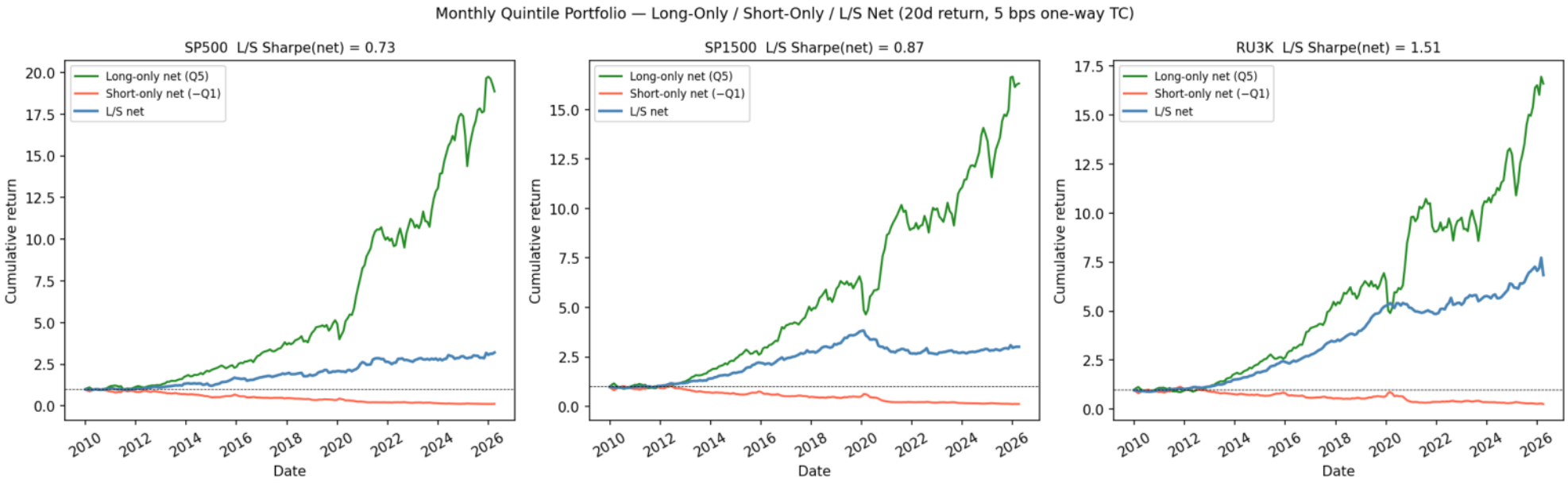
Spearman IC of ATCClassifierScore vs. forward returns (1–20d), by year and universe.
Consistent positive IC across 2010–2026; temporary dip in 2020 (COVID regime change).

§1b IC by GICS Sector



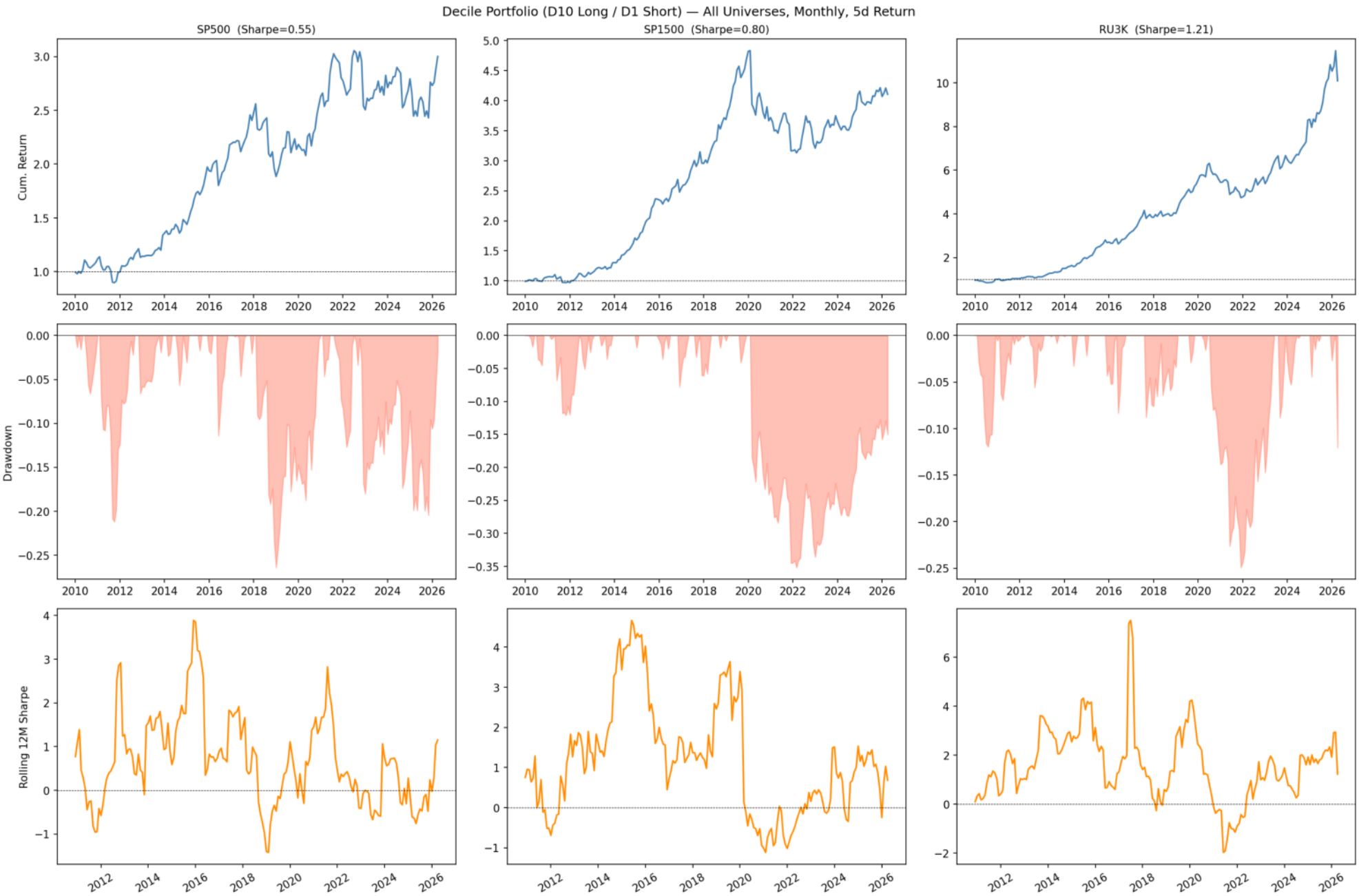
Spearman IC by sector (SP500, SP1500, RU3K). Consumer Staples and Energy lead;
Financials and Consumer Discretionary are weakest.

§2a Quintile Portfolio — Long / Short / L/S Equity Curves



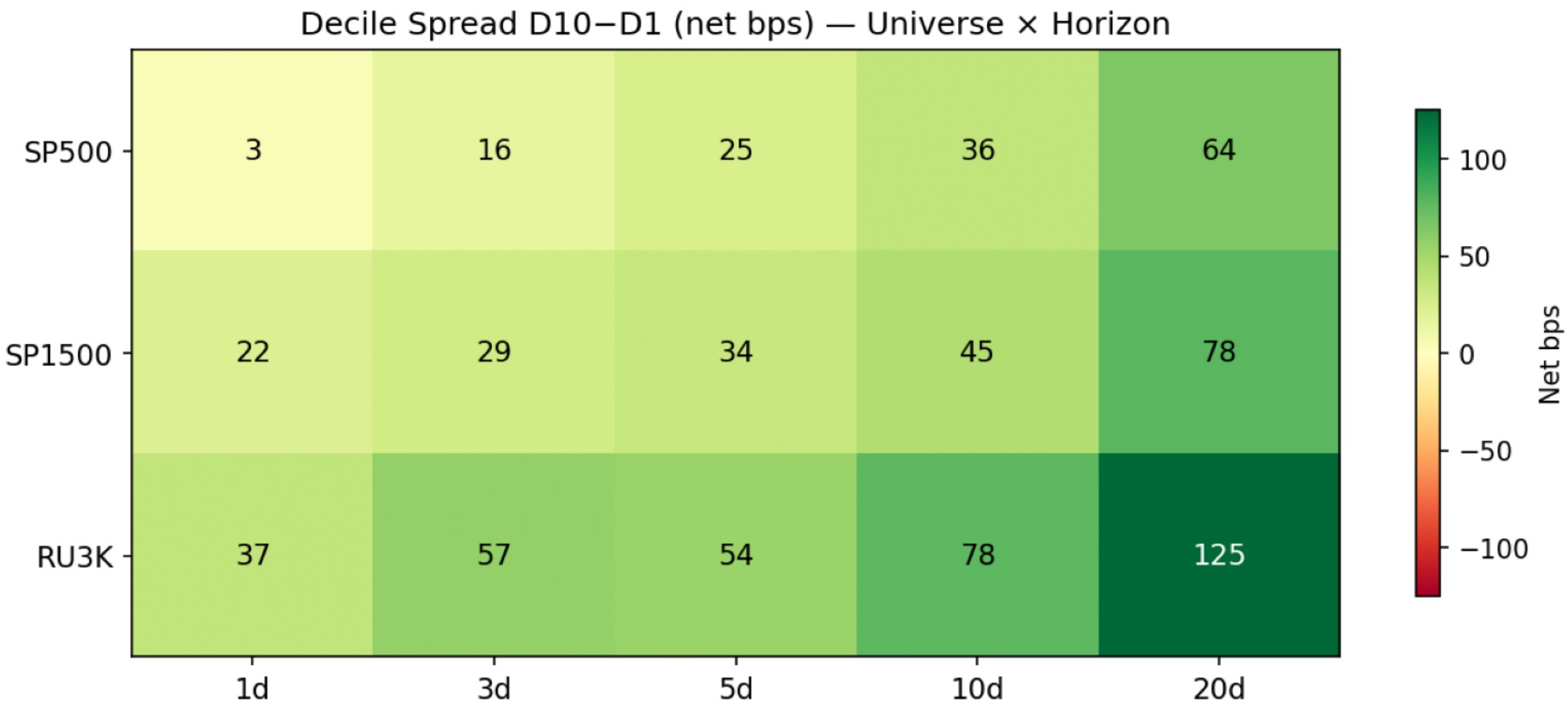
Monthly quintile L/S equity curves (20d holding, 5 bps one-way TC). Net Sharpe:
SP500 +0.73, SP1500 +0.87, RU3K +1.51.

§2b Decile Portfolio — Cumulative Return, Drawdown, Rolling Sharpe



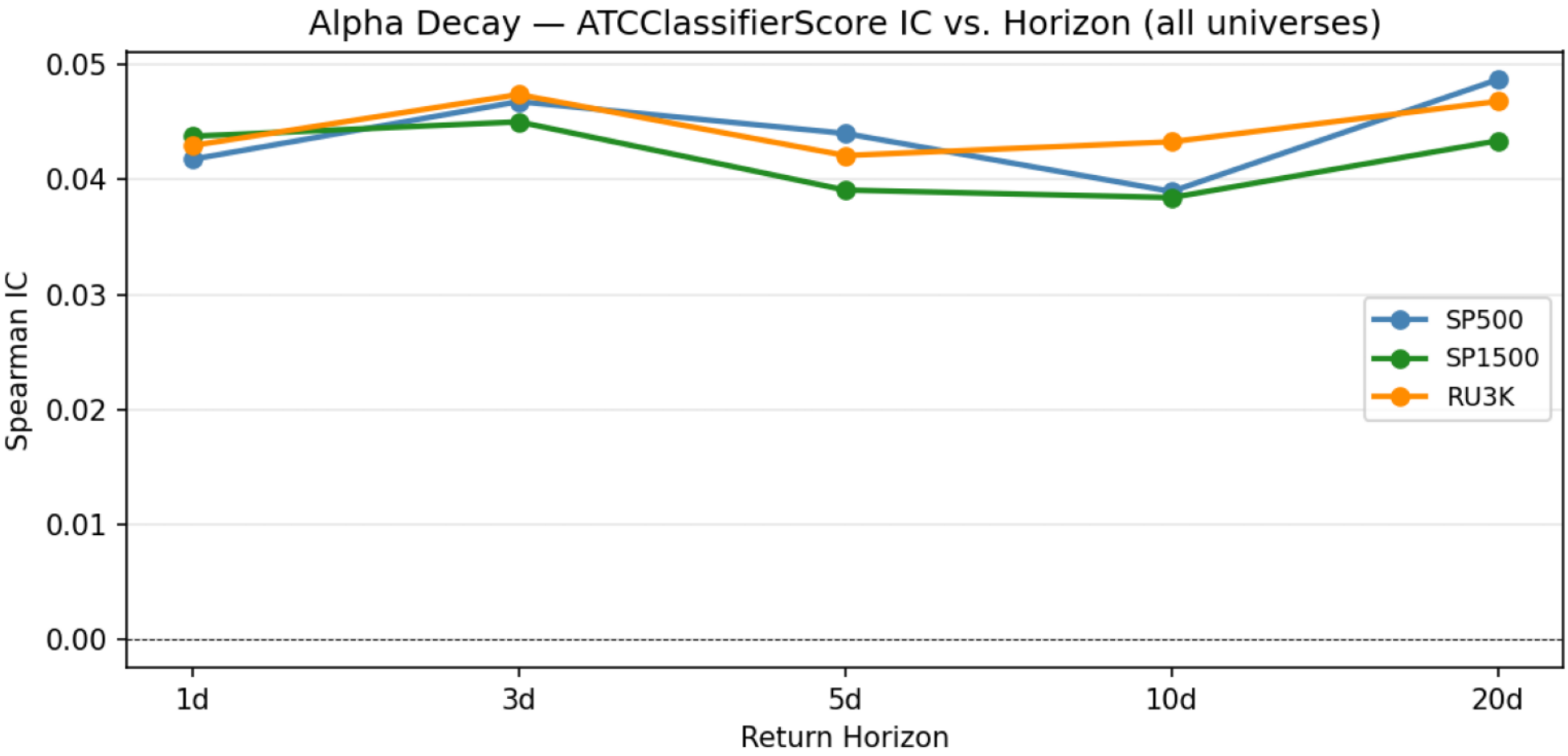
Top decile long / bottom decile short (D10−D1), monthly, 20d return, net of TC.
Rolling 12-month Sharpe and drawdown shown per universe.

§2b Decile Spread Heatmap (D10−D1 net, bps)



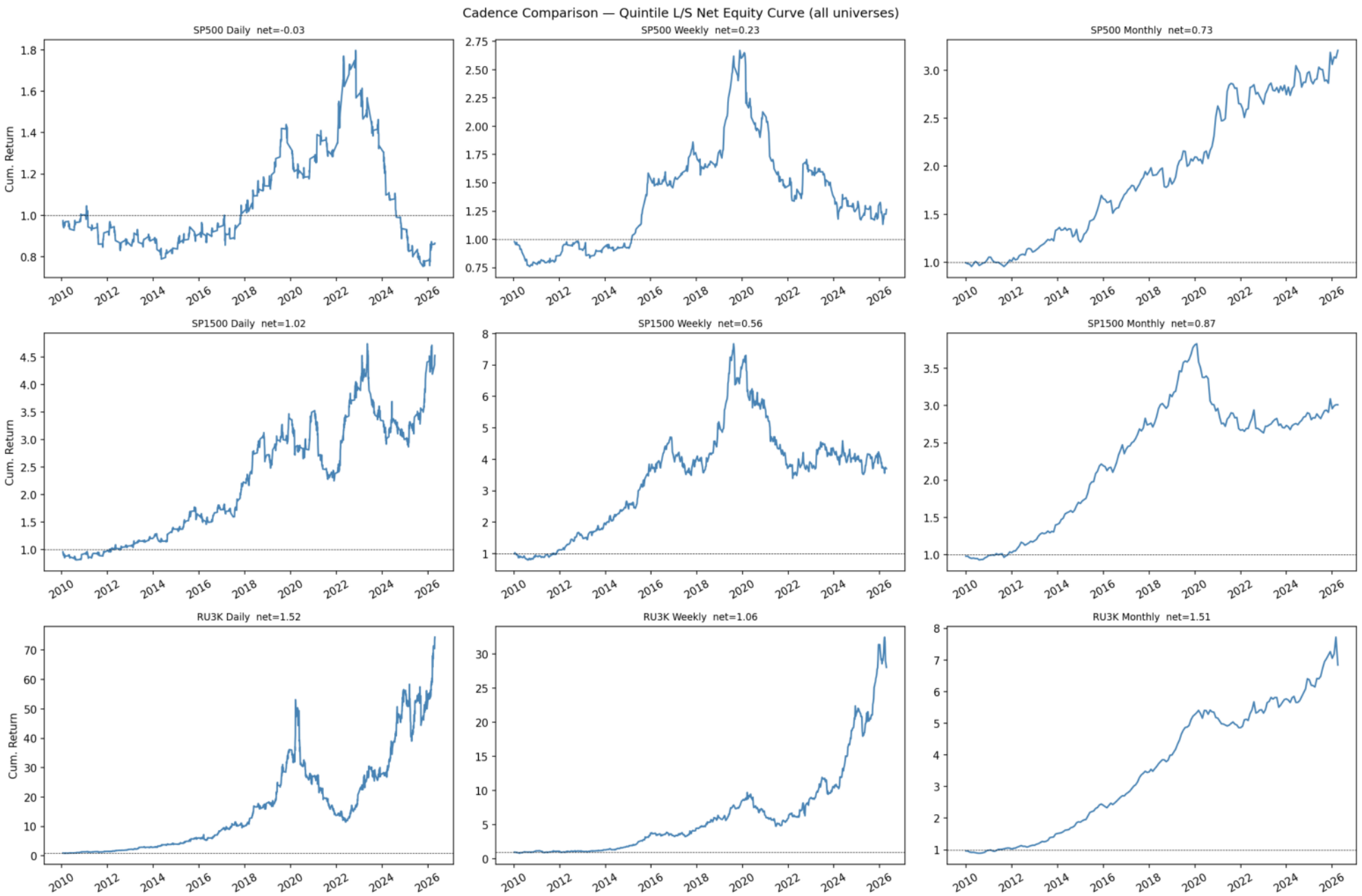
Net decile spread by universe × horizon. Spread grows monotonically from 1d to 20d, consistent with the ATC classifier's 14-day training window.

§2c Alpha Decay — IC vs. Return Horizon



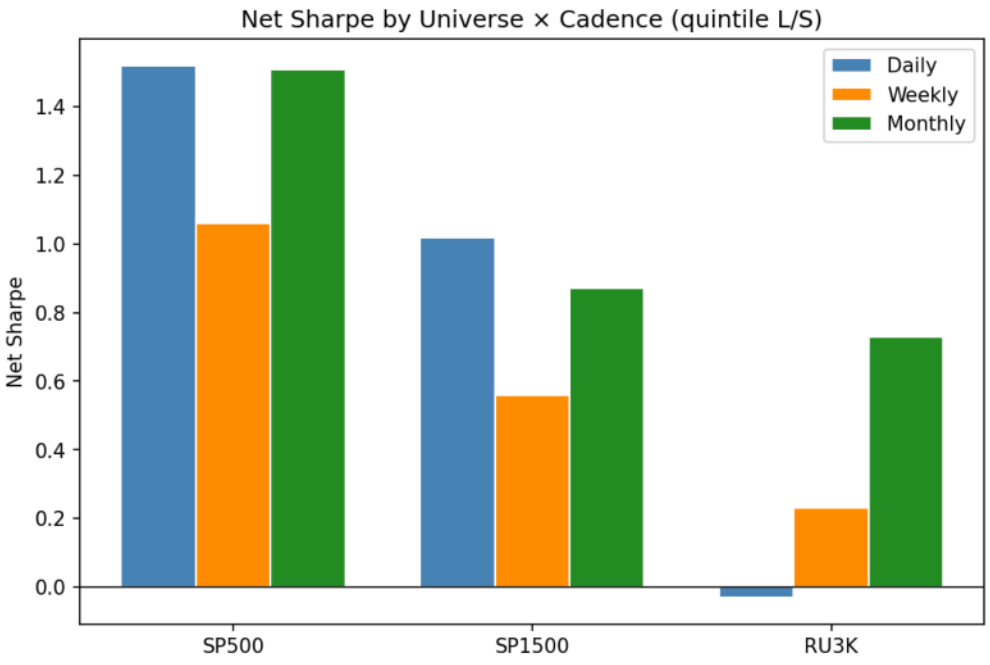
Spearman IC of ATCClassifierScore at 1, 3, 5, 10, 20-day horizons (all universes).
IC peaks at 10–20d, supporting monthly rebalancing as the primary cadence.

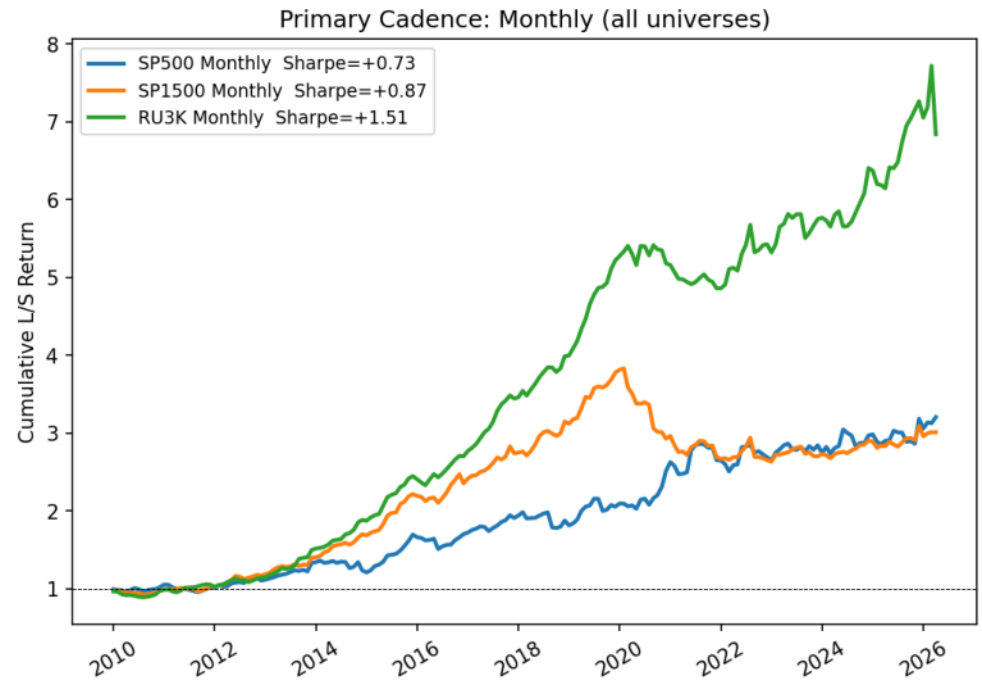
§2c Cadence Comparison — Daily / Weekly / Monthly



Quintile L/S cumulative equity curves across all cadences and universes.
Monthly is the only cadence with positive net Sharpe for all three universes.

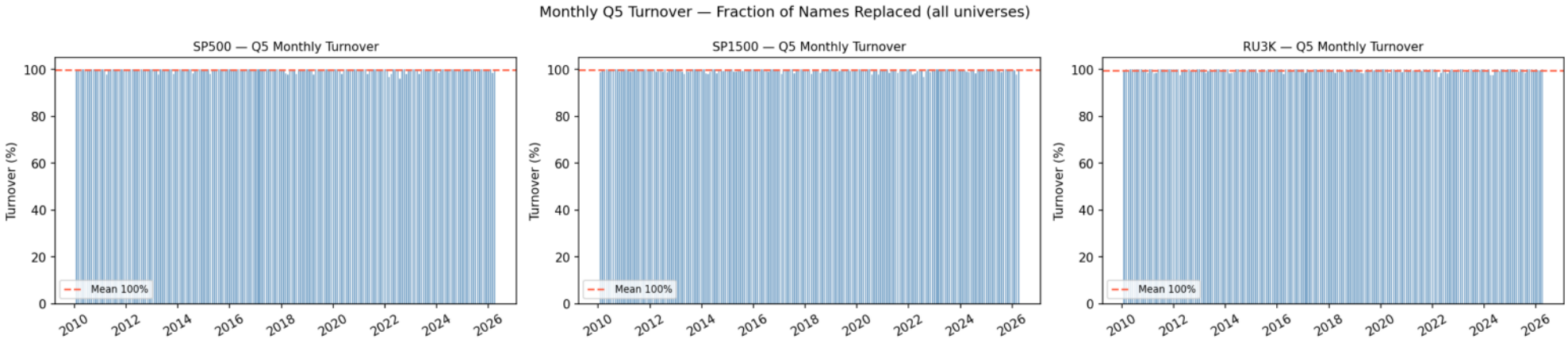
§2c Primary Cadence: Monthly (all universes)





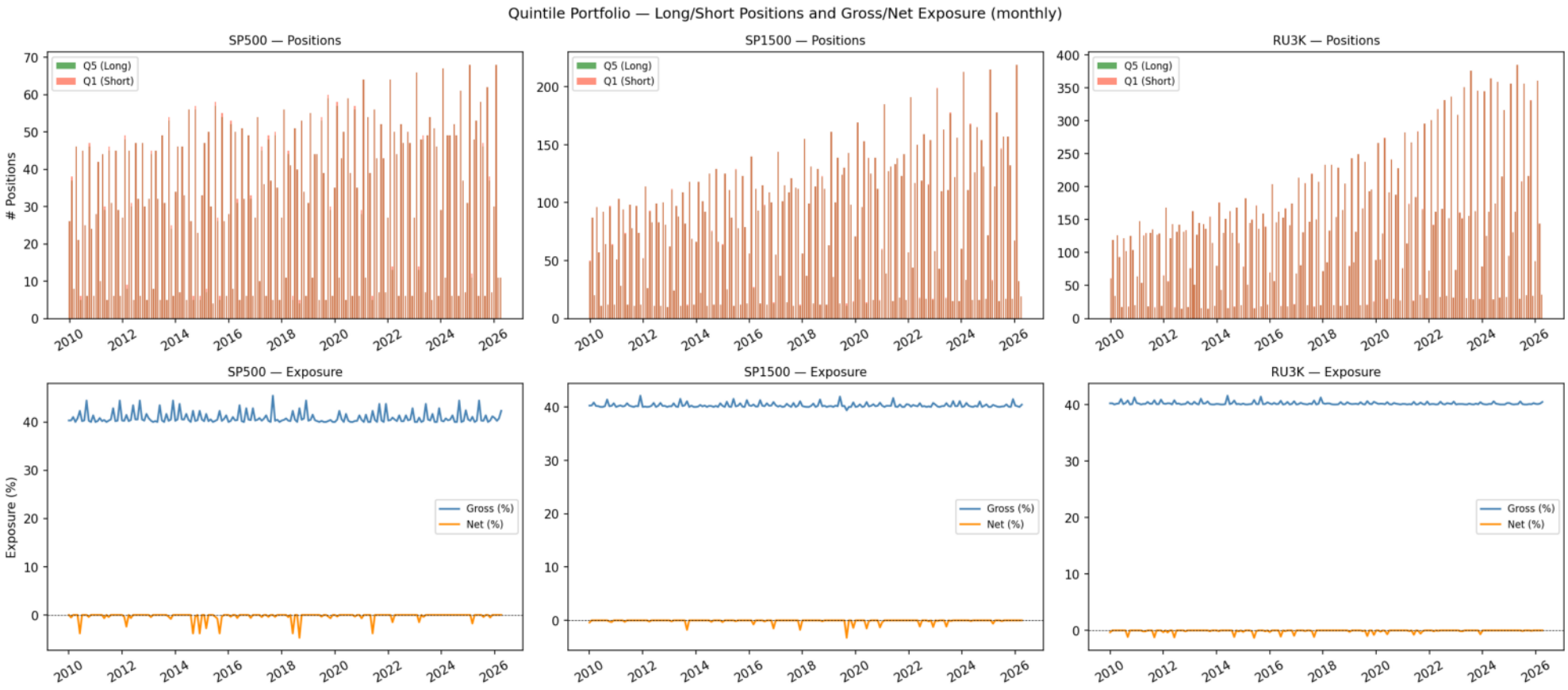
Composite equity curves using Monthly rebalancing. SP500 daily is TC-destroyed (net Sharpe −0.03); SP1500/RU3K daily shows higher drawdown risk.

§2d Turnover — Q5 Name-Level Churn



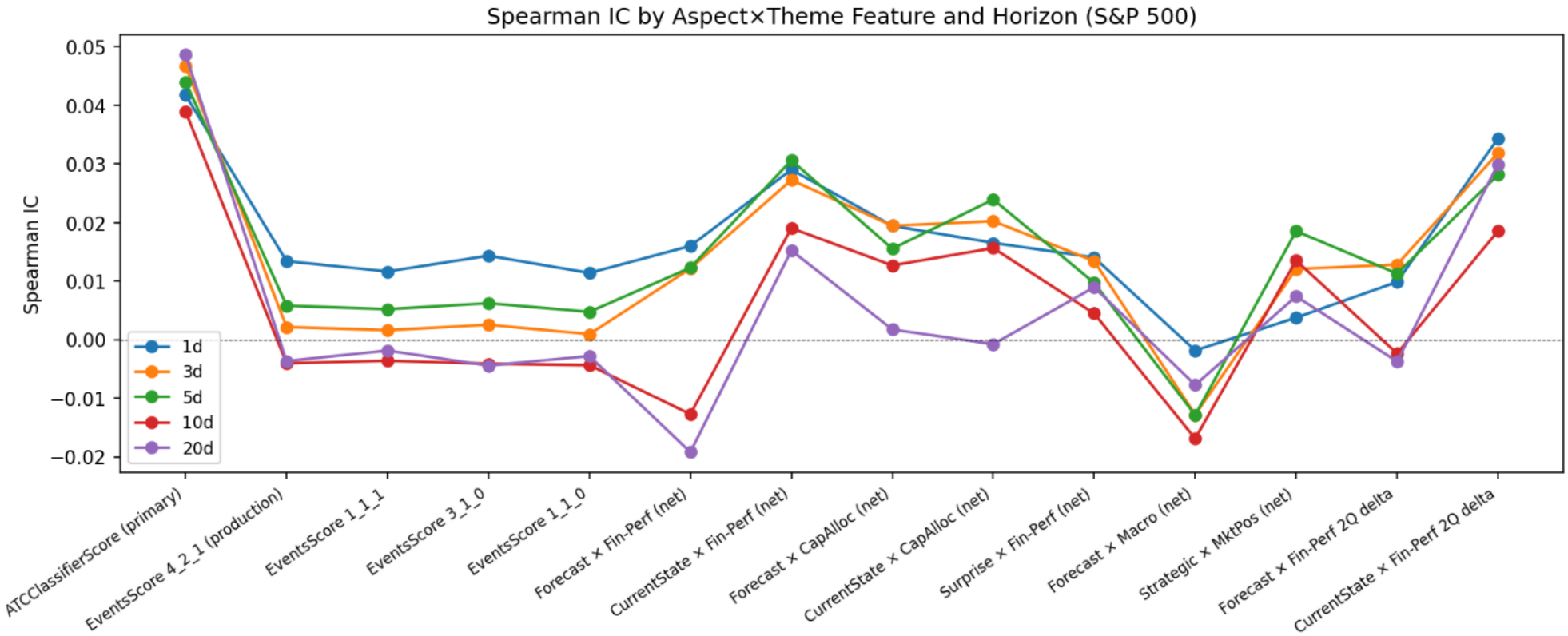
Fraction of Q5 (long) names replaced each month. Near-100% turnover confirms the full-turnover TC assumption used throughout.

§2e Gross / Net Exposure



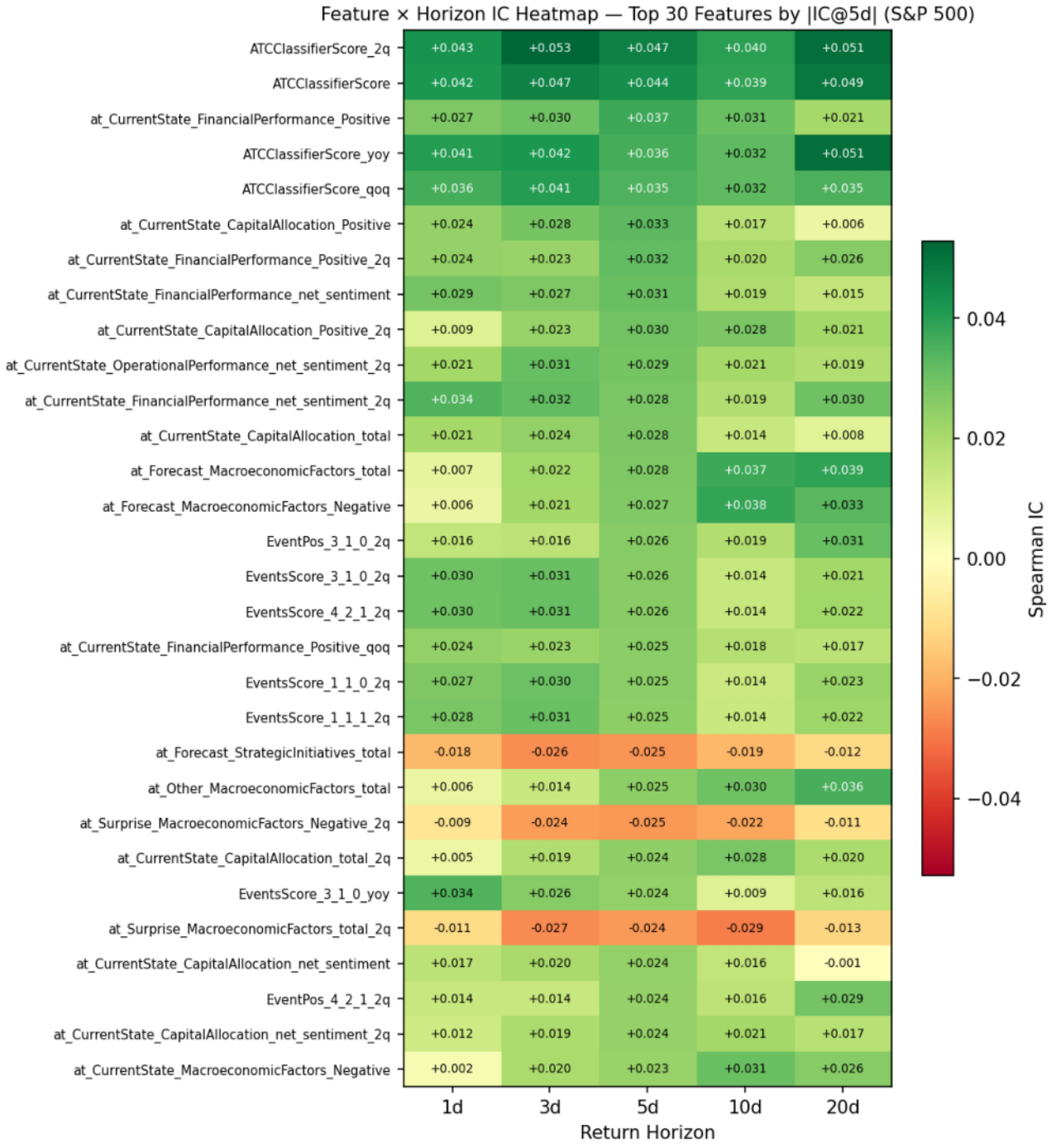
Equal-weight quintile construction produces an approximately dollar-neutral book.
Net exposure remains close to zero across the full sample.

§1c IC of Engineered Features (S&P 500)



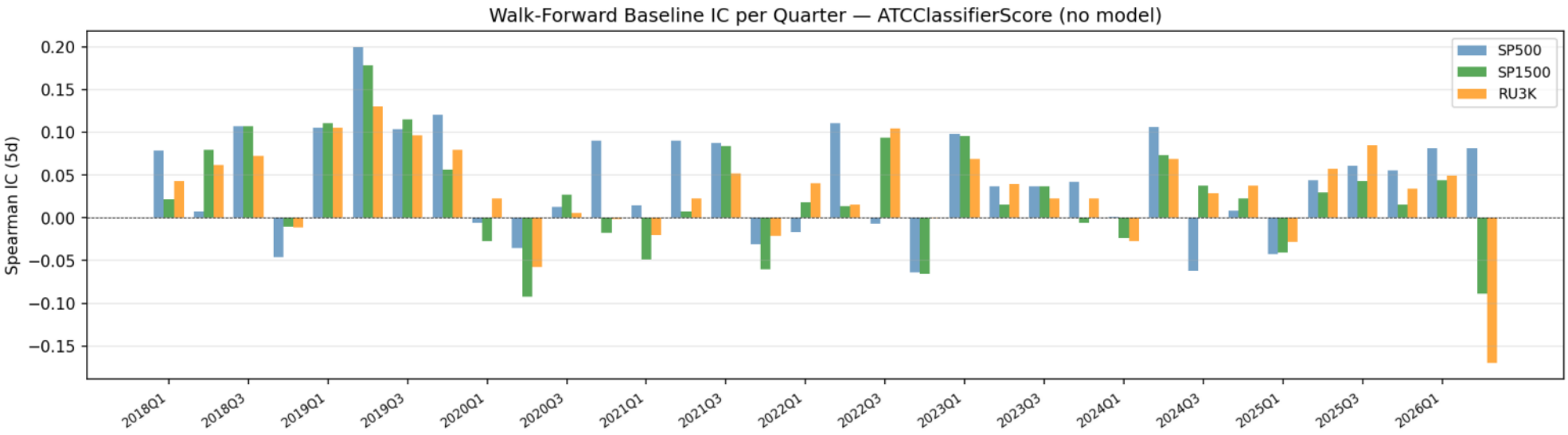
Spearman IC of key Aspect×Theme cross-product features vs. ATCClassifierScore baseline.
CurrentState×FinPerf 2Q delta and ATCClassifierScore_2q are the strongest individual features.

§1d Feature × Horizon IC Heatmap (Top 30 Features)



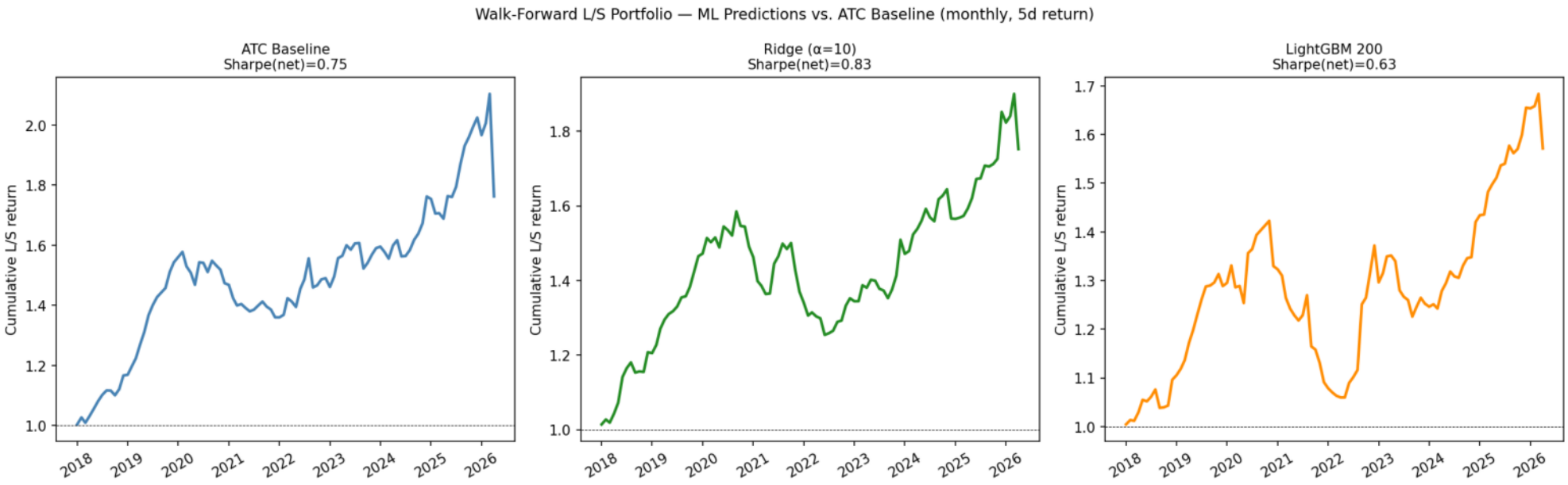
Top 30 engineered features ranked by |IC| at 5d (SP500). ATCClassifierScore and its trend variants (2q, QoQ, YoY) dominate across all horizons.

§2e Walk-Forward Baseline IC (ATCClassifierScore, no model)



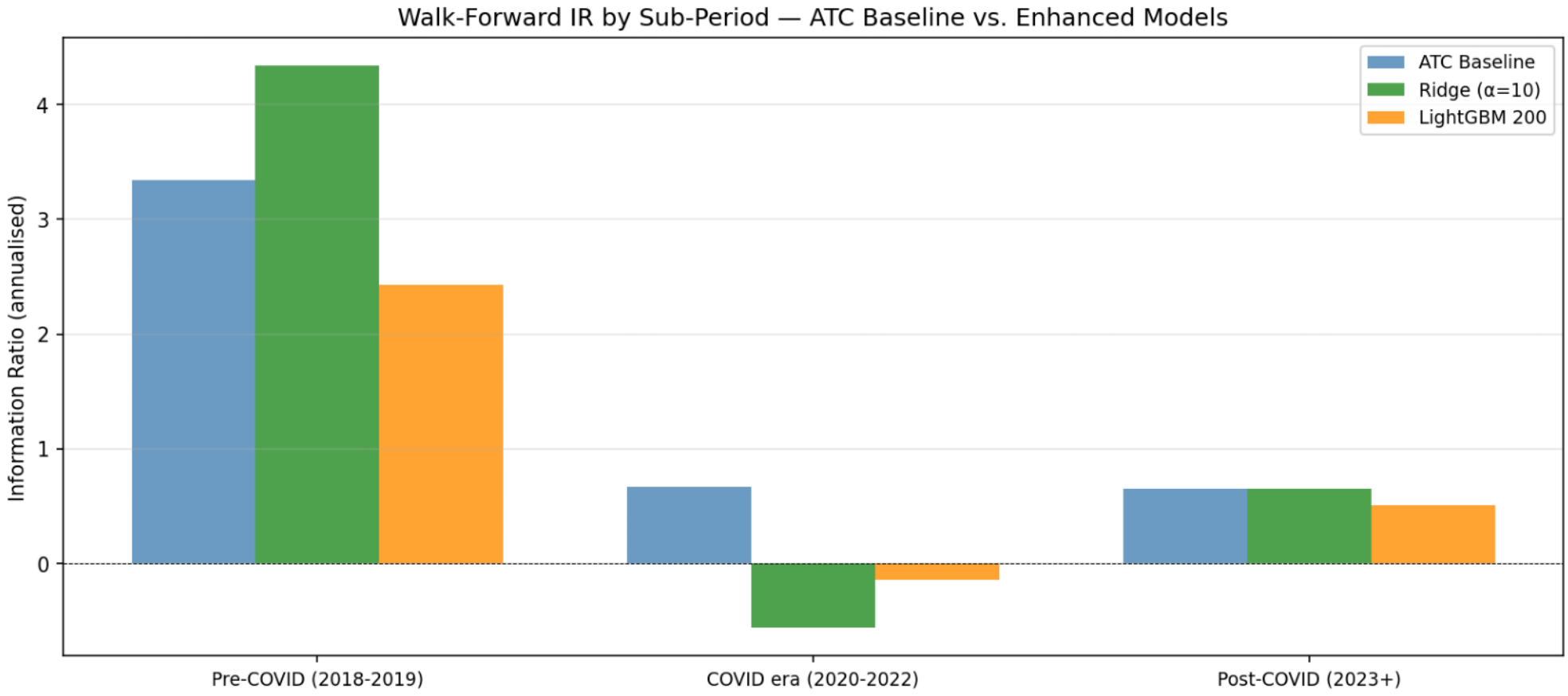
Quarterly out-of-sample Spearman IC, 2018Q1–2026Q2. Mean IC +0.030 (All universes), IR +1.09. Visible post-COVID decay from 2023 onward.

§4b Walk-Forward Portfolio — ATC Baseline vs. Ridge vs. LightGBM



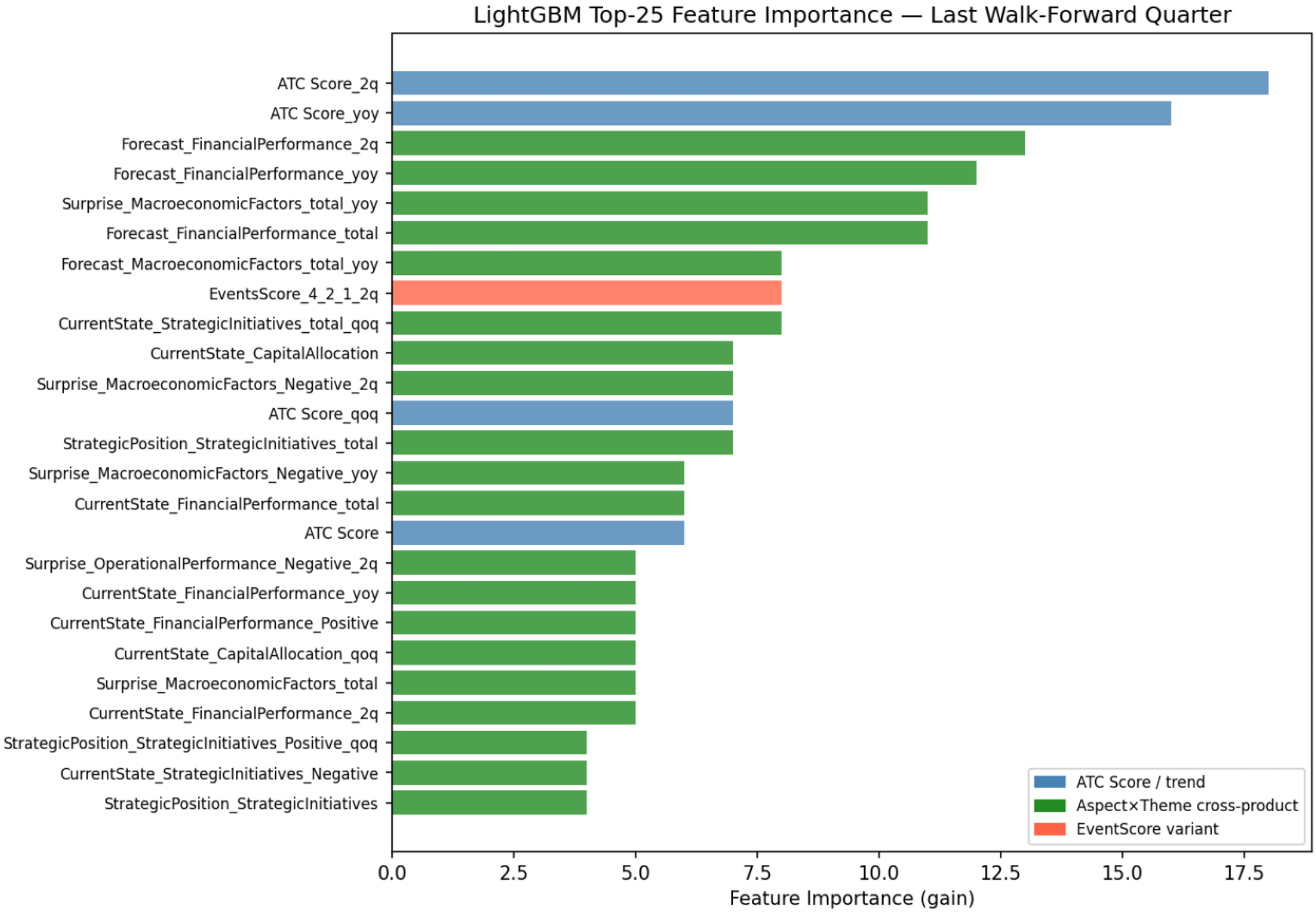
Monthly quintile L/S equity curves from OOS walk-forward predictions (all universes, 20 bps round-trip TC). Net Sharpe: Ridge +0.83, ATC Baseline +0.75, LightGBM +0.63.

§4d Walk-Forward IC — Sub-Period Breakdown



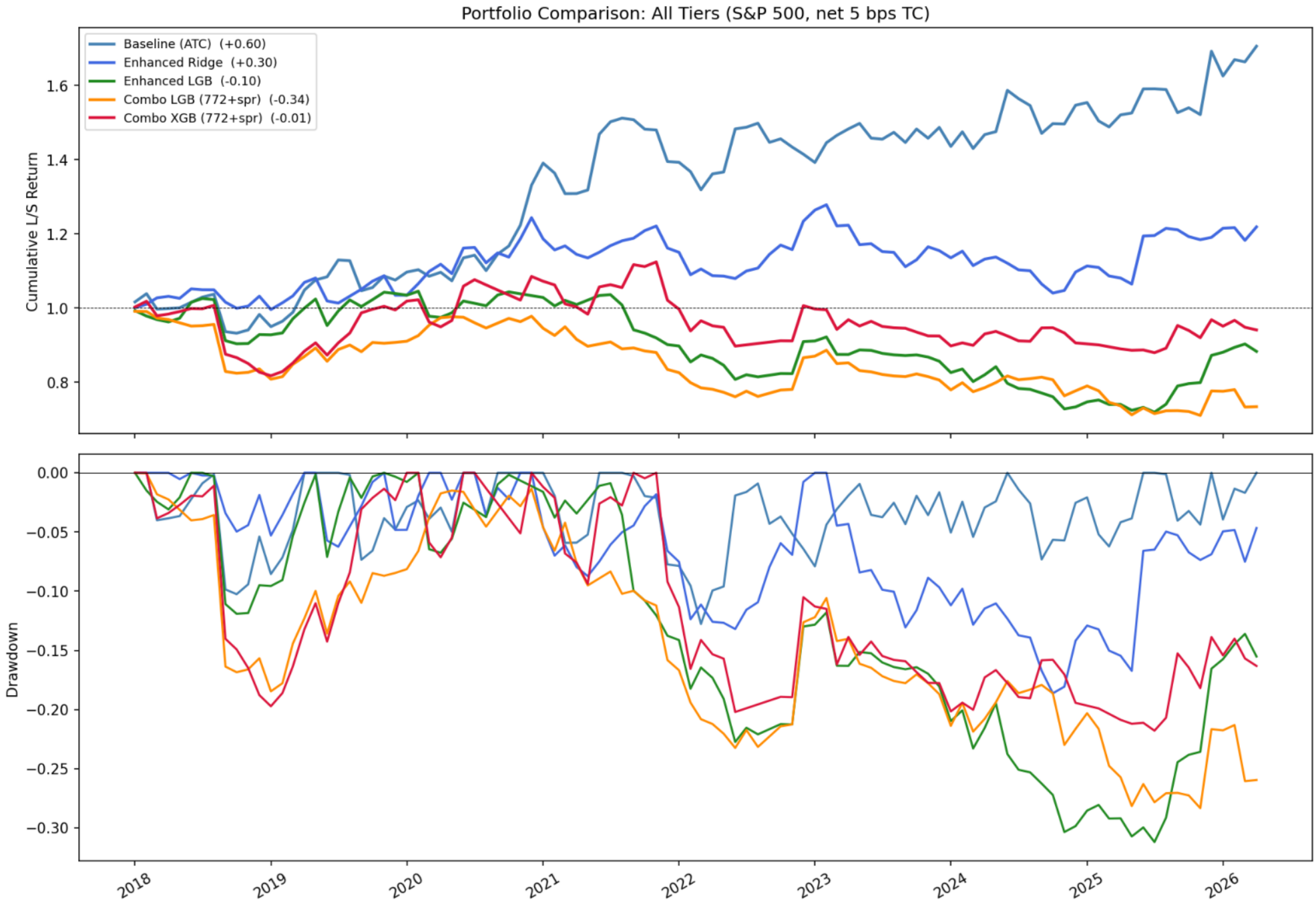
Information ratio by sub-period (Pre-COVID, COVID era, Post-COVID) for each model tier.
Post-COVID all models converge to IR +0.51–0.65; ML provides no additional resilience.

§4c LightGBM Feature Importance (Last Walk-Forward Quarter)



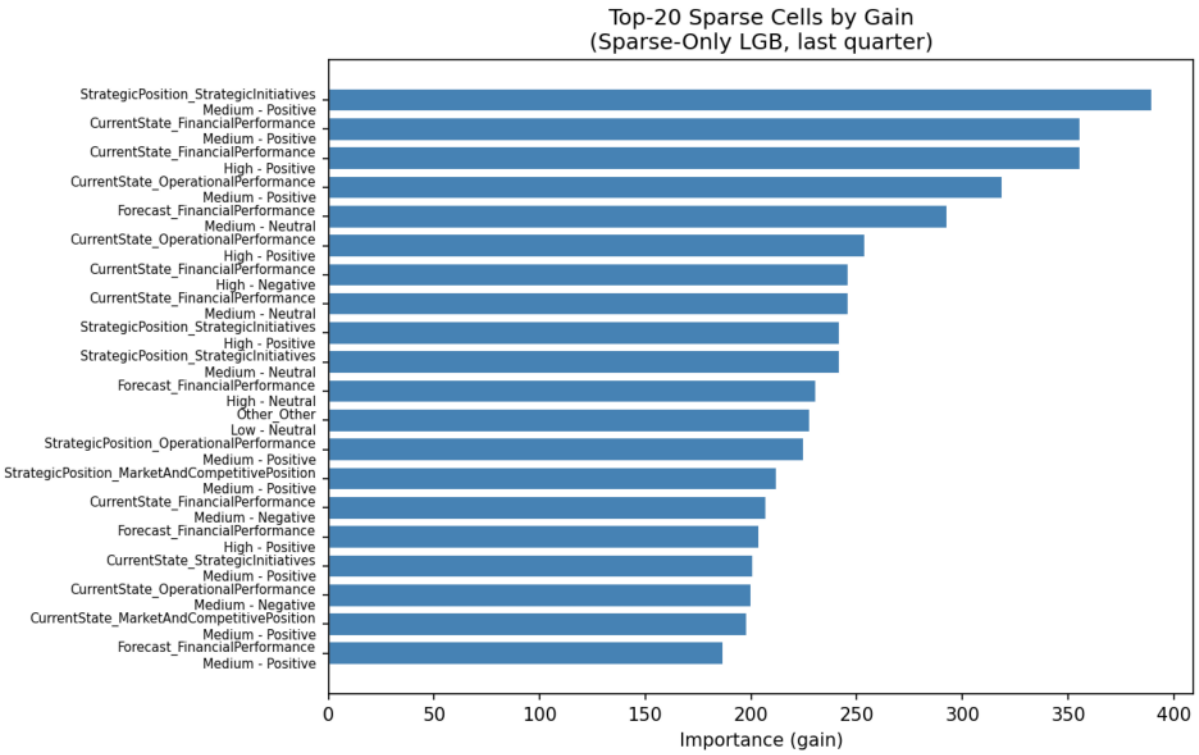
Top features by gain in the final walk-forward quarter. ATCClassifierScore trend variants and cross-product features dominate. Confirms 2q trend as the most valuable predictor.

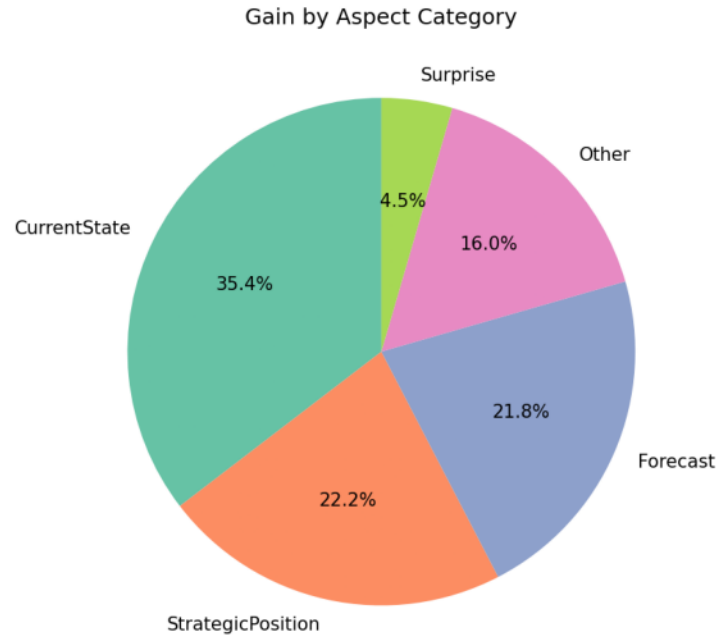
§3d Portfolio Comparison — All Tiers (SP500, Monthly)



OOS monthly quintile L/S equity curves: ATC Baseline, Enhanced Ridge/LGB,
Combo Ridge/LGB, Combo XGBoost. ATC Baseline (+0.60 Sharpe) is the strongest SP500 portfolio.

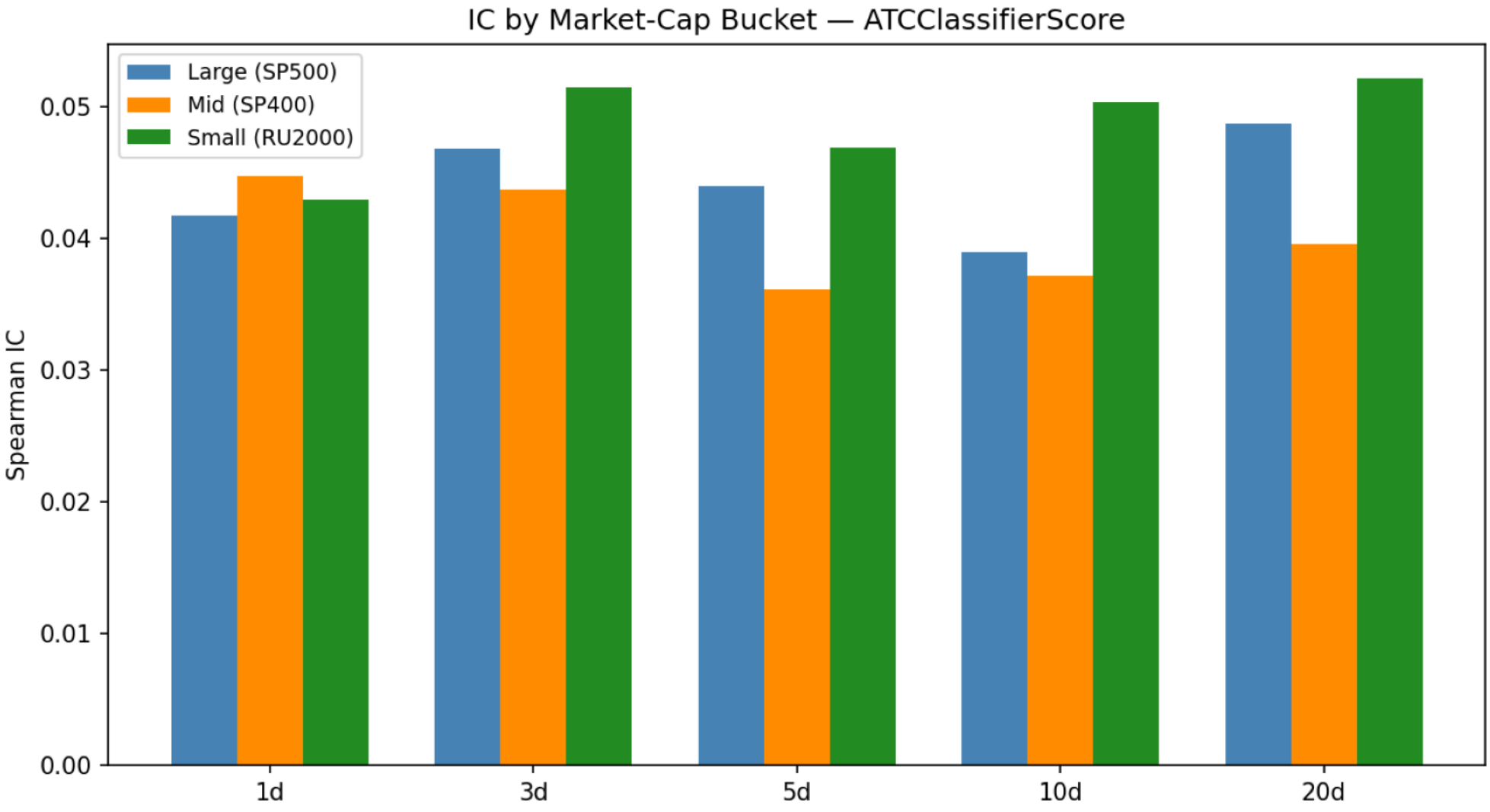
§3 Combo XGBoost Feature Importance





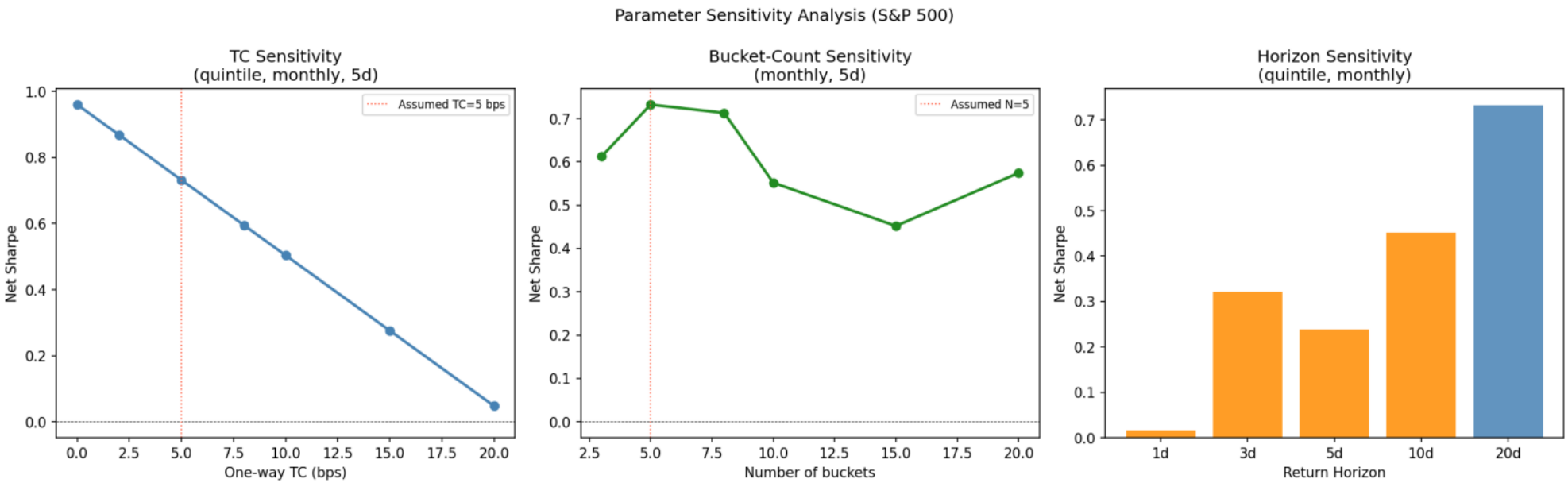
Top features in the Combo XGBoost model (772 engineered + 30 per-fold IC-selected sparse).
Sparse AspectTheme cells concentrated in positive FinPerf and CapAlloc language.

§5c IC by Market-Cap Bucket



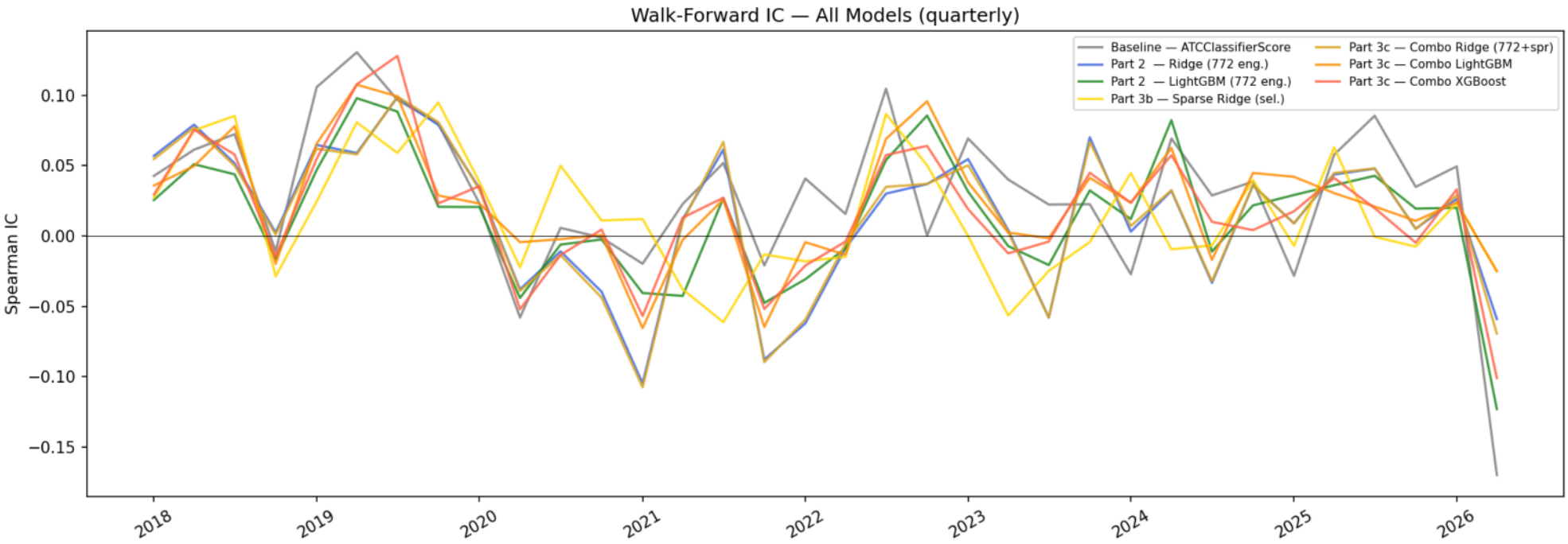
Spearman IC of ATCClassifierScore stratified by market-cap quartile (universe proxy).
Signal is consistent across size buckets — not exclusively a small-cap effect.

§5.6 Parameter Sensitivity — TC Level / Bucket Count / Return Horizon



(A) Break-even near ~20 bps one-way. (B) Quintile (5 buckets) near-optimal; octile adds no improvement.
(C) 20d holding period maximises net Sharpe (+0.73 vs +0.45 at 10d).

§5.3 Walk-Forward IC Summary — All Model Tiers



Quarterly OOS IC for all 7 model variants (2018Q1–2026Q2). Combo LightGBM IR +1.14 is the leading ML model; ATC baseline IR +1.09 serves as the benchmark floor.